

Mark Dostálík

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Experience

MSD Czech Republic

Prague, Czechia

DATA SCIENTIST / TECH LEAD

Jul. 2022 – present

- Leading development of software for processing high-resolution biological images and their downstream analysis. Semantic segmentation, image registration, statistics, software engineering.
- Tools & technologies: Python (NumPy, SciPy, Polars, OpenCV, scikit-learn, scikit-image, PyTorch, Pydantic, PyTest), AWS (EC2, S3), Git, CI/CD (GitHub Actions).

Mathematical Institute of Charles University

Prague, Czechia

POSTDOCTORAL RESEARCHER

Oct. 2021 – Jun. 2022

- Development of a mathematical model for cell membrane growth using the concepts from continuum mechanics and molecular dynamics. Solving systems of partial differential equations using finite element solvers.
- Tools & technologies: Python, FEniCS, Git.

Spaceknow, Inc.

Prague, Czechia

MACHINE LEARNING ENGINEER

Nov. 2016 – Sep. 2017

- Image processing of satellite imagery data. Object detection and instance segmentation using convolutional neural networks. Software development in Python. Running ML pipelines on premises as well as in cloud (Google Cloud Platform).
- Tools & technologies: Python, TensorFlow, Keras, Scikit-learn, Git.

Education

Faculty of Mathematics and Physics, Charles University

Prague, Czechia

PH.D. IN MATHEMATICAL AND COMPUTER MODELING

Oct. 2017 – Sep. 2021

- dynamical systems stability theory, partial differential equations, thermodynamics
- 6 journal publications, 1 proceedings publication
- principal investigator in a university research grant project

Skills

Programming

- Python (procedural and object-oriented), bash, basics of Rust
- Validation & testing: Pydantic, PyTest
- Data science packages: NumPy, SciPy, Polars/Pandas, scikit-learn, OpenCV, Plotly/Matplotlib
- Deep learning frameworks: PyTorch, Keras (Tensorflow)

Cloud & Infrastructure

- AWS (S3, EC2, EFS, Lambda)
- Google Cloud Platform

MLOps/DevOps

- MLflow
- Docker, mutagen
- CI/CD (GitHub Actions)

Languages

English

FULL PROFESSIONAL PROFICIENCY

Czech

NATIVE PROFICIENCY

Publications

Journals

- Allolio C., Fábíán B., Dostálík M.: OrganL: Dynamic Triangulation of Biomembranes using Curved Elements. *Biophys. J.*, 2024
- Boyaval S., Dostálík M.: Non-isothermal viscoelastic flows with conservation laws and relaxation, *J. Hyperbolic Differ. Equ.*, 2022
- Dostálík M., Průša V.: Non-linear stability and non-equilibrium thermodynamics—there and back again. *J. Non-equil. Thermodyn.*, 2022
- Dostálík M., Matyska C., Průša V.: Weakly nonlinear analysis of Rayleigh–Bénard convection problem in extended Boussinesq approximation. *Appl. Math. Comput.*, 408:126374, 2021
- Dostálík M., Průša V., Rajagopal K. R.: Unconditional finite amplitude stability of a fluid in a mechanically isolated vessel with spatially non-uniform wall temperature. *Contin. Mech. Thermodyn.*, 33:515–543, 2021
- Dostálík M., Průša V., Stein J.: Unconditional finite amplitude stability of a viscoelastic fluid in a mechanically isolated vessel with spatially non-uniform wall temperature. *Math. Comput. Simulat.*, 189:5–20, 2021
- Dostálík M., Málek J., Průša V., Süli E.: A simple construction of a thermodynamically consistent mathematical model for non-isothermal flows of dilute compressible polymeric fluids. *Fluids*, 5(3):133, 2020
- Dostálík M., Průša V., Tůma K.: Finite amplitude stability of internal steady flows of the Giesekus viscoelastic rate-type fluid. *Entropy*, 21(12), 2019

Proceedings

- Dostálík M., Průša V., Skřivan T.: On diffusive variants of some classical viscoelastic rate-type models. *AIP Conference Proceedings*, 2107(1):020002, 2019

Scientific projects

Charles University Grant Agency project

PRINCIPAL INVESTIGATOR

Prague, Czechia

Jan. 2019 – Sep. 2021

- Analysis of mathematical models of viscoelastic fluids with stress diffusion.
- Successfully fulfilled objectives of the project with 4 published results.

Scientific activities

Research stay at Laboratoire d'hydraulique Saint-Venant

VISITING RESEARCHER

Paris, France

Sep. 2020 – Dec. 2020

- Worked on mathematical modeling of flows of non-isothermal viscoelastic fluids.

International conferences

ORAL PRESENTER

- European Congress of Mathematics, Portorož, Slovenia (Jun. 2021)
- Multiscale Models for Complex Fluids: Modeling and Analysis, Banff, Canada (Nov. 2020)
- Bifurcations and Instabilities in Fluid Dynamics, Limerick, Ireland (Jul. 2019)
- Equadiff, Leiden, Netherlands (Jul. 2019)
- European Fluid Mechanics Conference, Vienna, Austria (Sep. 2018)

Honors & Awards

2016 **Winner**, 17th SVOČ competition: Applied Mathematics – mathematical modeling of dynamical systems

Brno, Czechia

Extracurricular Activity

SIAM Student Chapter Prague

VICE PRESIDENT IN 2018, 2019 & PRESIDENT IN 2020

Prague, Czechia

Jul. 2018 – Jun. 2021

- Acted as a core member in a student association of applied mathematicians.
- Organized student seminars, workshops, and several talks of top-tier scientists.

Undergraduate courses of calculus

TEACHER

Prague, Czechia

Oct. 2019 – Jun. 2020

- Taught 1st year undergraduate students of General Physics at Charles University.
- Homework and exams created for the course are available at <https://github.com/mark-dostalík/teaching>